

Lecture 15

Monday, 5/9/2022

Note Title

9/4/2022

Theorem 1: Let G be a group, and let $\{H_\alpha : \alpha \in \Delta\}$ be a collection of subgroups of G . Then, $\bigcap_{\alpha \in \Delta} H_\alpha$ is again a subgroup of G .

Proof: $e \in H_\alpha \quad \forall \alpha \in \Delta \Rightarrow e \in \bigcap_{\alpha \in \Delta} H_\alpha$.

Let $a, b \in \bigcap_{\alpha \in \Delta} H_\alpha$. Then, $a, b \in H_\alpha \quad \forall \alpha \in \Delta \Rightarrow ab^{-1} \in H_\alpha \quad \forall \alpha \in \Delta$
 $\Rightarrow ab^{-1} \in \bigcap_{\alpha \in \Delta} H_\alpha \quad \therefore \bigcap_{\alpha \in \Delta} H_\alpha \leq G$.

Union of subgroups need not be a subgroup.

Theorem 2: Let H and K be two subgroups of a group. Then, $H \cup K \leq G \Leftrightarrow$ either $H \subseteq K$ or $K \subseteq H$.

Proof: If $H \subseteq K$ or $K \subseteq H$, then $H \cup K = K$ or $H \cup K = H$.

$\therefore H \cup K$ is a subgroup of G .

Conversely, suppose that $H \not\subseteq G$, $K \not\subseteq G$ and $H \cup K \leq G$.

We prove that either $H \subseteq K$ or $K \subseteq H$.

If possible, suppose that $H \not\subseteq K$. Then, $\exists h \in H$ s.t. $h \notin K$.

Aim: $K \subseteq H$. Let $x \in K$. Then, $x, h \in H \cup K$.

$\Rightarrow xh \in H \cup K$ ($\because H \cup K \leq G$).

Case I: $xh \in H$. Then, $x = x1.h^{-1} \in H$.

Case II: $xh \in K$. Then, $h = x^{-1}.xh \in K$, a contradiction as $h \notin K$.

Thus, $x \in K \Rightarrow x \in H$. $\therefore K \subseteq H$. This completes the proof.

Ex: $U(8) = \{1, 3, 5, 7\}$. Let $H_1 = \{1, 3\}$, $H_2 = \{1, 5\}$, $H_3 = \{1, 7\}$.

Clearly, $H_i \not\subseteq H_j$ if $i \neq j$ and $U(8) = H_1 \cup H_2 \cup H_3$.

Hence, Theorem 3 is not true if we consider more than two subgroups. $\neq$

§ Subgroup generated by a subset X : Let G be a group and let X be a non-empty subset of G .

Let $\mathcal{H}(X) = \{H \leq G \mid X \subseteq H\}$ be the collection of all the subgroups of G containing X . Since $G \in \mathcal{H}(X)$, so $\mathcal{H}(X)$ is non-empty.

By Theorem 1, $\bigcap_{H \in \mathcal{H}(X)} H$ is a subgroup of G , which is clearly the smallest subgroup of G containing X .

Definition: Let X be a non-empty subset of G . The smallest subgroup of G containing X is called the subgroup generated by X , and is denoted by $\langle X \rangle$.

• If $X = \{a_1, \dots, a_n\}$ is a finite subset of G , then we write $\langle a_1, \dots, a_n \rangle$ in place of X .

• A group G is called finitely generated if $\exists a_1, \dots, a_n \in G$ such that $G = \langle a_1, \dots, a_n \rangle$.

Theorem 3: For a non-empty subset X of a group G , let $X^{-1} = \{x^{-1} : x \in X\}$. Then, the subgroup $\langle X \rangle$ consists of all the finite products of elements of $X \cup X^{-1}$.

That is, if $x \in \langle X \rangle$, then $x = x_1 * x_2 * \dots * x_n$ where $n \geq 1$ and

Proof: Let $F = \{x_1 * \dots * x_n \mid n \geq 1, x_i \in X \cup X^{-1} \forall i\}$.

Claim: $F = \langle X \rangle$.

Clearly, F is a subgroup of G and $X \subseteq F$. Since $\langle X \rangle$ is the smallest subgroup of G containing X , so $\langle X \rangle \subseteq F$. Let H be a subgroup of G s.t. $X \subseteq H$. Then, $X^{-1} \cup X \subseteq H \Rightarrow F \subseteq H$.
 $\Rightarrow F \subseteq \langle X \rangle$, by taking $H = \langle X \rangle$. Hence, $F = \langle X \rangle$. $\#$

Ex: $SL_2(\mathbb{Z}) = \langle \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \rangle$. [we will prove it later]

Let $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

$$\therefore SL_2(\mathbb{Z}) = \left\{ T^{k_1} S^{k_2} \cdots T^{k_{n-1}} S^{k_n} \mid n \geq 1, k_i \in \mathbb{Z} \forall i \right\}.$$

Here, we apply Theorem 3 with $X = \{T, S\}$ in the group $SL_2(\mathbb{Z})$.

§ Commutativity measure:

- Let G be a cyclic group. Then, $G = \langle a \rangle$.

Let $x, y \in G$. Then, $x = a^n$ and $y = a^m$ for some $n, m \in \mathbb{Z}$.

$$\therefore xy = a^{n+m} = yx.$$

This proves that every cyclic group is abelian.

Definition: Let G be a group, and $a \in G$. The centralizer of 'a' in G is the set $C_G(a) = \{x \in G \mid xa = ax\}$.

- For every $a \in G$, $C_G(a)$ is a subgroup of G .

Definition (Centre of a group): Let G be a group. Then, the centre of G is defined as

$$Z(G) = \{x \in G \mid xy = yx \quad \forall y \in G\}.$$

- $Z(G)$ is a subgroup of G .
- G is abelian $\Leftrightarrow Z(G) = G$.
- $Z(G) = \bigcap_{a \in G} C_G(a)$.

Ex: Find $Z(Q_8)$ and $Z(GL_2(\mathbb{R}))$.
